SHARESAFE

SOFTWARE REQUIREMENTS SPECIFICATION

A browser-based image privacy and preparation toolkit

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Status: Draft for product and engineering review

WORKING PRODUCT NAME

ShareSafe

Branding and domain name remain subject to confirmation.

Document Control

Field	Value
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Product name	ShareSafe (working title; final brand/domain to be confirmed)
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Primary delivery target	Public web application / MVP
Intended audience	Product owner, UX/UI designer, frontend engineer, QA engineer, deployment owner

Revision History

Version	Date	Description	Author / Owner
1.0	11 Nov 2026	Initial SRS based on the agreed MVP concept and discussed feature set.	Project team

Purpose of This Document

This document defines the product scope, externally observable behavior, functional requirements, quality attributes, technical constraints, acceptance criteria and release boundaries for ShareSafe. It is intended to be the shared reference for design, implementation, testing and deployment.

Naming note

ShareSafe is used as the working product name throughout this SRS. BlurBefore was discussed as an alternative brand idea. The final public name and domain must be selected and checked before launch.

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1. Introduction

1.1 Purpose

ShareSafe enables people to prepare images for safer sharing by allowing them to manually obscure sensitive areas, remove common embedded metadata through re-encoding, resize images and compress image files. The product is designed to be quick to use, accessible without an account and privacy-conscious by processing images locally in the user's browser.

1.2 Intended Audience

This SRS is intended for the product owner, frontend developers, UX/UI designers, quality assurance engineers, and anyone responsible for deployment, support or product review.

1.3 Product Objective

Provide a simple, globally accessible tool that reduces accidental exposure of personal or confidential information when users share screenshots and photos online. The MVP should solve the main task without requiring server-side image processing, user registration or a paid third-party API.

1.4 Scope of This Specification

The document covers the initial public MVP and identifies optional capabilities that are explicitly deferred. It does not specify a backend image-processing service, account system, cloud image storage or automated cloud-based image analysis.

2. Product Overview

2.1 Product Description

ShareSafe is a responsive web application. A user selects an image from their device, marks areas that should be hidden, optionally changes dimensions or compression quality, removes supported metadata by exporting a newly encoded image, previews the result and downloads it. The original image is not uploaded to ShareSafe servers as part of the standard workflow.

2.2 Problem Statement

People routinely share screenshots, receipts, documents, support conversations and photos. These files can visibly expose phone numbers, email addresses, account identifiers, private messages or other sensitive content. Image files may also contain metadata such as GPS coordinates or capture details. General-purpose image editors can be excessive for a quick privacy task, and users may not know whether a simple crop or overlay actually removes the underlying content.

2.3 Product Value

- Fast manual redaction without installing desktop software.
- Privacy-first processing: the image is handled locally in the browser.
- Flattened export so redaction marks are burned into the output pixels rather than remaining as editable overlays.
- Practical image preparation features, including metadata cleaning, resizing and compression.
- No account or sign-up requirement for the core workflow.

3. Users and Operating Environment

3.1 Primary User Groups

User group	Typical task	Expected benefit
Everyday social media user	Hide details in a screenshot before posting.	Quick, low-friction privacy protection.
Remote worker / support user	Redact account details or private messages before sharing a support screenshot.	Reduce accidental disclosure.
Freelancer / small business user	Prepare receipts, invoices or customer screenshots for sharing.	Share a useful image without visible confidential details.
Student / educator	Hide names, IDs or personal information in educational screenshots.	Safer sharing of learning materials.

3.2 User Characteristics

Users may have limited image-editing experience. They may use a desktop, laptop, tablet or mobile browser. The application must provide clear selection handles, visible feedback, readable controls and a predictable download action.

3.3 Operating Environment

- Modern desktop and mobile browsers with support for the File API, Canvas 2D, Blob URLs and standard image decoding.
- Initial target browsers: current stable versions of Chrome, Edge, Firefox and Safari. Exact minimum versions are to be set during implementation and verified in QA.
- Internet access is required to load the website. After the application assets have loaded, image editing and export should not require an application-server request.
- Processing performance depends on the user's device memory, CPU, browser and source-image dimensions.

4. Product Scope and Release Boundaries

4.1 MVP In Scope

ID	Capability	MVP commitment
SCP-01	Image intake	Select an image using a file picker; drag-and-drop may be supported as an equivalent entry path.
SCP-02	Manual redaction	Apply blur and opaque solid-cover redactions to user-selected rectangular regions.
SCP-03	Preview	Display the image and the redaction changes before export.
SCP-04	Metadata cleaning	Re-encode supported image formats to produce an output without common source EXIF/GPS metadata; verify supported formats.
SCP-05	Resize	Set output width/height while preserving aspect ratio by default.
SCP-06	Compression	Choose an output format and quality setting where the format supports quality control.

ID	Capability	MVP commitment
SCP-07	Download	Export a flattened image file to the user's device.
SCP-08	Privacy and accessibility	No sign-up, no image upload, responsive UI, clear errors and privacy explanation.

4.2 Explicitly Out of Scope for MVP

- User accounts, authentication, cloud storage, saved projects or cross-device editing.
- Uploading user images to an Express/API server for processing.
- Automatic OCR detection of emails, phone numbers, account IDs or passwords.
- Automatic face detection or face blurring.
- Bulk/batch processing of multiple images in one operation.
- Collaboration, sharing links, team workspaces or image hosting.
- Native mobile applications, browser extensions or desktop applications.
- Guaranteeing that all possible metadata types are removed from every image format and every browser implementation.

MVP scope guardrail

The first release is a manual image editor and export utility. Automatic detection and batch processing are future enhancements, not launch blockers.

5. User Journeys and Use Cases

5.1 Primary User Journey

1. User opens the ShareSafe home page.
2. User selects or drops one supported image into the upload area.
3. The browser reads and decodes the image locally and opens the editor.
4. User selects a redaction tool (blur or solid cover) and marks one or more rectangular regions.
5. User reviews the preview and adjusts or removes any incorrect redaction regions.
6. User optionally selects resize dimensions, output format and compression quality.
7. User chooses Export / Download. The application renders a flattened image and re-encodes it.
8. The browser starts a local download. The application releases temporary processing resources when they are no longer required.

5.2 Use Case Catalogue

Use case	Actor	Precondition	Successful outcome
UC-01 Upload image	Visitor	Home page is loaded.	A supported image is displayed in the editor.
UC-02 Blur a region	Visitor	An image is open.	Selected region is visibly blurred in preview and export.
UC-03 Cover a region	Visitor	An image is open.	Selected region is covered by an opaque fill in preview and export.
UC-04 Remove metadata	Visitor	Supported image is loaded.	Exported file is re-encoded and common source metadata is not retained.

Use case	Actor	Precondition	Successful outcome
UC-05 Resize image	Visitor	An image is open.	Export uses requested dimensions, subject to configured limits.
UC-06 Compress image	Visitor	An image is open.	Export is generated at selected format/quality and a resulting size is shown when available.
UC-07 Download image	Visitor	Export is complete.	File is saved through the browser download flow.

6. Functional Requirements

The requirements below are uniquely identified and written so that they can be verified by inspection or test. Priority values: Must = required for MVP; Should = desirable for MVP if time permits; Could = optional.

6.1 Landing Page and Entry

ID	Priority	Requirement
FR-001	Must	The system shall provide a public landing page explaining the purpose of ShareSafe and the supported privacy workflow.
FR-002	Must	The landing page shall provide a clearly visible action to select or drop an image.
FR-003	Must	The landing page shall state that image processing is performed locally in the browser and that images are not uploaded for normal processing.
FR-004	Should	The landing page shall include concise explanations of redaction, metadata cleaning, resizing and compression.
FR-005	Should	The website shall include a privacy explanation, supported-format information and a help/FAQ section.

6.2 Image Upload and Validation

ID	Priority	Requirement
FR-006	Must	The system shall allow the user to select a single image file from the local device.
FR-007	Should	The system shall support drag-and-drop as an alternative to the file picker.
FR-008	Must	The system shall validate file type and reject unsupported or non-image files with a readable message.
FR-009	Must	The system shall validate file size and image dimensions against configurable client-side limits before processing.
FR-010	Must	The system shall display a loading or processing state while a large image is being decoded.
FR-011	Must	The system shall not transmit the selected image or its pixel data to ShareSafe application servers.
FR-012	Should	The system may allow the user to paste an image from the clipboard where browser permissions and APIs allow it.

6.3 Editing and Redaction

ID	Priority	Requirement
FR-013	Must	The editor shall display the loaded image at a usable scale while preserving its original aspect ratio.
FR-014	Must	The user shall be able to select a rectangular region on the image for redaction.
FR-015	Must	The user shall be able to apply a blur effect to each selected region.
FR-016	Must	The user shall be able to apply an opaque solid-color cover to each selected region.
FR-017	Must	The editor shall visually distinguish the current selection from already-applied redactions.
FR-018	Must	The user shall be able to move, resize or remove a redaction region before export.
FR-019	Must	The editor shall support multiple redaction regions on one image.
FR-020	Should	The editor shall provide undo, redo and reset-to-original actions for editing operations.
FR-021	Should	The editor shall provide zoom controls or equivalent navigation for precision when editing large images.
FR-022	Must	Redaction effects shall be rendered into the exported image pixels; the export shall not contain editable overlay objects that reveal the original pixels beneath a redaction.

6.4 Preview and Export

ID	Priority	Requirement
FR-023	Must	The system shall provide a preview of the edited image before the user downloads it.
FR-024	Must	The export process shall render the original image and all accepted edits into a new flattened raster output.
FR-025	Must	The system shall provide a download action for the generated file.
FR-026	Must	The downloaded file shall use a meaningful filename with an output suffix, such as "-sharesafe".
FR-027	Must	The system shall show a clear success or failure state for export.
FR-028	Should	The system shall display the output file size and dimensions after export when those values can be measured reliably.

6.5 Metadata Cleaning

ID	Priority	Requirement
FR-029	Must	The export pipeline shall re-encode the supported input image instead of copying the source file byte-for-byte.
FR-030	Must	For supported formats, the export shall remove common source EXIF metadata, including GPS location fields, by decoding and re-encoding the image.

ID	Priority	Requirement
FR-031	Must	The UI shall explain that metadata cleaning applies to supported metadata in the exported file and is not a guarantee against every possible metadata or steganographic payload.
FR-032	Must	The team shall verify metadata-removal behavior for every input/output format and target browser combination included in the release support matrix.
FR-033	Should	The system shall label metadata cleaning as enabled by default for all standard exports.

6.6 Resize and Compression

ID	Priority	Requirement
FR-034	Must	The user shall be able to specify an output width and/or height within configured limits.
FR-035	Must	Aspect ratio shall be preserved by default when the user changes one dimension.
FR-036	Should	The user shall be able to unlock aspect ratio when intentionally choosing independent width and height values.
FR-037	Must	The user shall be able to choose among supported export formats, initially JPEG, PNG and WebP where supported by the browser.
FR-038	Must	For lossy output formats, the user shall be able to select a compression-quality level.
FR-039	Must	The system shall communicate that JPEG/WebP quality settings may reduce visual quality and that PNG is generally lossless but may create larger files.
FR-040	Should	The system shall offer a small number of understandable quality presets such as Smaller file, Balanced and Higher quality.

6.7 Local State and Resource Cleanup

ID	Priority	Requirement
FR-041	Must	The application shall hold working image data in browser memory only for the active session and shall not persist it to application storage.
FR-042	Must	The application shall release object URLs, canvas buffers and other temporary resources when an image is replaced, the editor is reset or the page is closed, where browser APIs permit.
FR-043	Must	Refreshing or closing the page may discard the current edit; the user shall be informed that projects are not saved.
FR-044	Should	The editor shall warn before discarding unsaved work when the user starts a new image or resets the current project.
FR-045	Must	When exporting a transparent image to JPEG, the system shall flatten transparent pixels onto a defined background (white by default) and communicate that transparency is not supported by JPEG.

6.8 Optional Client-Side Detection (Future)

Automatic OCR detection of emails, phone numbers, account identifiers and automatic face detection were discussed as potential enhancements. They are not required for the MVP. If implemented later, processing should remain local to the browser by default, and detected regions must be presented for user review before they are redacted.

7. User Interface Requirements

7.1 Core Screens

Screen	Required elements
Home / landing	Product explanation, privacy promise, upload area, supported file formats, primary call to action, FAQ and privacy link.
Editor	Image canvas, zoom/navigation controls, blur tool, solid-cover tool, region controls, undo/redo/reset, export settings and download action.
Export state	Processing indicator, success/failure message, output dimensions/size where available, download action.
Privacy / FAQ	Plain-language explanation of local processing, metadata limits, redaction best practices, supported formats and troubleshooting.

7.2 Editor Interaction Requirements

- Tool selection must be visually obvious; the active tool must be indicated.
- The cursor or touch interaction must clearly indicate when the user is creating or manipulating a redaction rectangle.
- Redaction region handles must be large enough to operate on touch screens.
- The editor must distinguish between blur and opaque cover so users understand which method is active.
- The export button must be disabled while an export is already running to prevent duplicate processing.
- All critical actions must have accessible labels and keyboard focus states.

7.3 Responsive Design

The application shall support desktop, tablet and mobile viewport widths. On smaller screens, controls may move into a bottom toolbar, collapsible panel or stacked layout. The image canvas must remain usable without horizontal page overflow.

8. Data and Processing Requirements

8.1 Data Classification

Data	Handling rule	Persistence
Source image bytes	Read locally through browser file APIs; do not upload.	In-memory for active session only.
Redaction coordinates and settings	Maintain in browser state for the active editing session.	Not persisted in MVP.
Generated output	Create locally as Blob/object URL and download through browser.	Saved by user to their own device.

Data	Handling rule	Persistence
Website assets	Served from static hosting/CDN.	Hosted deployment assets.
Operational telemetry	Optional privacy-conscious page-level metrics only; never include image contents or sensitive pixel data.	Only if separately configured and disclosed.

8.2 Client-Side Processing Flow

Step	Processing behavior
1. Read	Browser receives a local File object from the user.
2. Validate	Check MIME/type, file size, dimensions and whether the browser can decode the image.
3. Decode	Decode the image into a browser-supported image representation.
4. Edit	Render the image and user-selected redaction regions in the editor; keep editing controls separate from the source pixels.
5. Flatten	Draw the final image and all redactions into a new canvas or equivalent raster surface.
6. Transform	Apply optional resize and selected output format/quality.
7. Re-encode	Create a new output Blob. This is the metadata-cleaning boundary for supported formats.
8. Download	Create a temporary object URL and trigger a browser download; release temporary resources afterward.

8.3 Supported Formats for Initial Release

Format	Input	Output	Notes
JPEG / JPG	Must	Must	Lossy output; quality control should be available.
PNG	Must	Must	Suitable for screenshots and transparency; export behavior must be tested.
WebP	Should	Should	Support depends on target browser; verify browser encoding support.
GIF	Out of MVP	Out of MVP	Animated images require frame-aware handling; do not silently export only one frame.
HEIC / HEIF, TIFF, SVG	Out of MVP	Out of MVP	May require additional decoders or security handling; reject clearly in MVP.

Important implementation detail

A visual blur layer or HTML overlay is not sufficient for safe redaction. The final export must be a flattened raster image in which the redaction pixels replace the original pixels in the selected areas.

9. Non-Functional Requirements

9.1 Privacy and Security

ID	Requirement
NFR-001	The MVP shall not send image files, pixel data, redaction coordinates or generated output to ShareSafe servers.
NFR-002	The application shall not use third-party image-processing APIs or upload-based OCR/face-detection services.
NFR-003	The application shall use HTTPS in production.
NFR-004	The application shall avoid logging file names, image metadata, redaction content or other image-derived personal information.
NFR-005	The application shall clearly disclose any optional analytics or third-party scripts that are loaded.
NFR-006	The export pipeline shall flatten edits and must not retain the original unredacted pixels beneath redacted areas in the exported file.
NFR-007	The application shall use a Content Security Policy and reasonable security headers appropriate for a static web application.

9.2 Performance

ID	Requirement
NFR-008	The landing page should load quickly on a typical broadband connection; avoid large unnecessary frontend dependencies.
NFR-009	For images within the published supported size limit on a representative modern device, the editor should become interactive within 5 seconds after file selection.
NFR-010	The interface shall provide visible progress or busy feedback for operations that take longer than 500 ms.
NFR-011	Large-image operations should be moved off the main UI thread using Web Workers and OffscreenCanvas where practical and supported, with a tested fallback.
NFR-012	The application shall define and enforce a configurable maximum file size and maximum decoded pixel dimension to reduce memory exhaustion risk.

9.3 Usability and Accessibility

ID	Requirement
NFR-013	Primary tasks shall be discoverable without account creation or onboarding.
NFR-014	The application shall provide clear validation, processing and error messages in plain language.
NFR-015	Text, controls and focus indicators shall meet WCAG 2.2 AA targets where applicable.
NFR-016	The interface shall support keyboard navigation for primary controls; canvas interactions shall have an accessible alternative or clear documented limitation. Color alone shall not communicate tool state or errors.

9.4 Compatibility and Reliability

ID	Requirement
NFR-017	The application shall be tested against the agreed current stable browser matrix for Chrome, Edge, Firefox and Safari.
NFR-018	Unsupported formats, insufficient memory, decode failures and export failures shall be handled without crashing the page.
NFR-019	The site shall remain usable on mobile screens and common touch input devices.
NFR-020	The application shall not depend on a long-running application server for core editing functionality.

9.5 SEO and Discoverability

ID	Requirement
NFR-021	Each public page shall have a unique title, meta description, canonical URL and meaningful heading structure.
NFR-022	The landing page shall explain the tool in indexable text rather than relying only on text embedded in images.
NFR-023	The site shall provide a sitemap.xml, robots.txt, Open Graph metadata and a clean URL structure.
NFR-024	Search landing pages for distinct use cases may be added, but must offer genuinely useful content and must not be thin or duplicated doorway pages.

10. Technical Architecture and Deployment

10.1 Recommended Architecture

Layer	Technology / approach	Responsibility
Web application	Next.js with static-friendly deployment	Landing page, editor UI, help/privacy pages and client-side application logic.
Styling	Tailwind CSS	Responsive UI and consistent design system.
Image manipulation	HTML Canvas 2D and browser image APIs	Draw image, redaction regions, resize and export.
Background work	Web Workers; OffscreenCanvas where practical	Move heavy processing away from the UI thread.
File handling	File API, Blob, URL.createObjectURL	Read local files and produce downloadable output.

10.2 Logical Data Flow

Component	Communicates with	Purpose
Browser UI	Browser image APIs	Select, display and edit the local image.
Editor state	Canvas rendering layer	Maintain redaction coordinates and user settings for the active session.
Export module	Canvas / Blob APIs	Flatten and re-encode the final image.
Static hosting	Browser	Deliver HTML, JavaScript, CSS, fonts and static assets only.
Optional analytics	Analytics provider, if enabled	Record aggregate page usage; no image or image-derived data.

10.3 Backend Decision

No dedicated backend in MVP

An Express server, PostgreSQL database, Redis instance, object storage and paid image API are not required for the core product. The host still serves the website and may receive normal HTTP requests for static assets. The user's image-processing workload runs on the user's device.

10.4 Deployment Requirements

- Deploy the Next.js site to a static-capable hosting platform with HTTPS enabled.
- Configure production domain, DNS, redirects and canonical host consistently.
- Use environment variables only for non-secret build/deployment configuration; no user image credentials or server secrets are needed for MVP.
- Configure basic uptime and deployment checks. If analytics are enabled, document the provider and data collected.
- Keep the site functional if optional analytics are blocked by browser privacy settings.

11. Error Handling and Edge Cases

Scenario	Expected system behavior
Unsupported file type	Reject the file and list supported formats.
File exceeds size or pixel limit	Do not attempt expensive processing; explain the limit and suggest a smaller image.
Image cannot be decoded	Show a readable error and allow the user to choose another file.
Browser lacks requested output format	Disable or omit that format and offer a supported alternative.
Export fails or browser runs out of memory	Preserve the current editor state where possible, show an error and allow retry with smaller dimensions.
User replaces image	Ask for confirmation if edits would be discarded; release old temporary resources.
User closes or refreshes tab	Current work may be lost; no server-side recovery is available in MVP.
Redaction area is too small or outside bounds	Clamp coordinates to the image and provide visible selection feedback.
Mobile touch interaction	Support touch-based selection and keep controls reachable.
Metadata removal limitations	Display the documented scope and avoid claiming universal metadata or forensic sanitization.

12. Acceptance Criteria

The MVP is acceptable for public beta when the following criteria are met:

- A visitor can open the site and use the editor without signing in or creating an account.
- A visitor can select at least JPEG, PNG and supported WebP images within the published file and dimension limits.
- The visitor can create multiple blur regions and multiple opaque cover regions on one image.
- Redaction regions can be adjusted or removed before export.
- The exported image visibly contains the redactions as flattened pixels; reopening the export does not reveal the original content underneath those areas.
- The export process re-encodes the image and a test suite confirms that common EXIF/GPS metadata is absent for supported format/browser combinations.
- The visitor can resize an image with aspect ratio preservation enabled by default.
- The visitor can choose supported output formats and lossy quality settings where applicable.
- The visitor can download the result and receives a meaningful success or error message.
- Network inspection confirms that source image bytes and generated output are not sent to ShareSafe application servers during normal editing and export.
- The application handles unsupported files, large images, decode errors and export failures without an unrecoverable page crash.
- The editor is usable on supported desktop and mobile browsers, with no critical layout overflow.
- The privacy page clearly explains local processing, metadata-removal limits and the distinction between blur and opaque redaction.

13. Test Strategy

13.1 Functional Testing

- Upload each supported file format and reject unsupported formats.
- Create, move, resize, delete and combine multiple redaction regions.
- Export images with blur only, cover only and mixed redactions.
- Verify dimensions, aspect ratio, output format, quality settings and downloaded filename.
- Reopen exported files in an independent image viewer to verify the result.

13.2 Privacy and Security Testing

- Use browser developer tools/network capture to confirm image files and pixel data are not uploaded.
- Inspect exported files with metadata inspection tools to verify expected metadata removal for each supported format/browser.
- Confirm redactions are burned into output pixels and no original pixels can be recovered from the exported file by removing a UI layer.
- Check that console logs, analytics events and error telemetry do not contain image contents or sensitive image-derived values.
- Validate HTTPS and production security headers.

13.3 Performance and Compatibility Testing

- Test representative small, medium and near-limit images on a range of device capabilities.
- Test browser memory pressure and graceful failure behavior.
- Test latest stable Chrome, Edge, Firefox and Safari, plus mobile Safari and Chrome where available.
- Test keyboard-only navigation, focus visibility, touch targets and screen-reader labels for primary controls.

13.4 Suggested Test Image Set

Fixture	Purpose
JPEG with EXIF and GPS sample metadata	Verify that supported metadata is not retained after export.
PNG screenshot with text	Verify crisp solid-cover redaction and image dimensions.
WebP photo	Verify decode, edit and output behavior.
Very large high-resolution image	Verify size/pixel limits and performance handling.
Transparent PNG	Verify whether transparency is retained for PNG output and handled correctly for JPEG output.
Corrupted and renamed non-image files	Verify robust input validation.

14. Risks, Assumptions and Dependencies

14.1 Risks

Risk	Impact	Mitigation
Browser memory limits on large images	Tab may become slow or crash.	Set clear limits; use workers; test large files; show graceful errors.

Risk	Impact	Mitigation
Metadata behavior varies by format/browser	Some metadata may remain or format support may differ.	Define supported matrix; re-encode; inspect outputs in automated QA; state limitations.
Users confuse visual blur with secure redaction	Sensitive content could remain recoverable in a non-flattened output.	Flatten redactions; emphasize opaque cover for highly sensitive information; test exported pixels.
Free hosting tier limits change	Unexpected hosting cost or throttling.	Keep deployment portable and monitor provider quotas.
Low search visibility after launch	Few organic visitors initially.	Create genuinely useful search pages, publish demos and collect feedback; do not assume immediate traffic.
Domain/brand conflicts	Rebranding or domain replacement may be needed.	Check domain availability, trademark considerations and social handles before purchase.

14.2 Assumptions

- The first release is a public web utility rather than a native app.
- The product will use a working brand name until a final name/domain is confirmed.
- The user has a modern browser with standard local file and canvas capabilities.
- No image is retained by ShareSafe after the browser session; downloads are saved by the user.
- The initial user workflow is one image at a time.

14.3 Dependencies

- Next.js and selected frontend dependencies.
- Browser support for required image, canvas and file APIs.
- Static-capable hosting provider and domain registrar.
- Test fixtures and metadata inspection tools for validation.

15. Future Enhancements

The following features may be considered after the MVP has been launched and user demand has been validated. They are not commitments for the first release.

Enhancement	Description	Implementation note
Automatic sensitive-text detection	Detect emails, phone numbers and selected identifiers.	Prefer browser-local OCR; require user review before applying redaction.
Face detection and blur	Suggest face regions for blurring.	Use an on-device model where feasible; disclose model download/performance impact.
Batch processing	Process multiple images in one session.	Assess memory, device limits and UX before implementation.
Additional formats	Consider GIF frame-aware support, HEIC and TIFF.	Only add when decoding/export semantics and privacy are verified.
Preset redaction styles	Saved style presets for common workflows.	Keep settings local unless a sync/account feature is justified.

Enhancement	Description	Implementation note
PWA/offline support	Allow the tool shell to load as an installable/offline-capable app.	Cache static application assets; do not cache user images by default.
Privacy-focused analytics	Understand page-level traffic and tool engagement.	Aggregate metrics only; no image content or sensitive derived data.
Premium batch workflow	Optional paid convenience features.	Requires business validation and a payment/security review; core privacy workflow should remain useful for free.

16. Glossary and Open Decisions

16.1 Glossary

Term	Meaning
Redaction	Permanently obscuring selected image content in the exported raster output.
Blur	A visual effect that reduces legibility. For high-sensitivity content, an opaque cover is safer.
Flattening	Rendering the image and edits into one raster output so editing overlays are not separate removable layers.
EXIF	A metadata standard commonly used to store camera, timestamp and sometimes GPS information in image files.
Canvas API	A browser API used to draw and manipulate raster images.
Blob	A browser object representing raw file-like data, used here to create an export.
Web Worker	A browser execution context that can run processing away from the main UI thread.
MVP	Minimum viable product: the smallest release that delivers the core user value.

16.2 Open Decisions Before Development

Decision	Recommended starting point	Owner / status
Final brand and domain	Use ShareSafe as internal working name; confirm final name and domain after availability/trademark checks.	Product owner - Open
Maximum source image size	Start with a configurable 20-25 MB cap, then validate on target devices.	Engineering - To confirm
Maximum decoded dimensions	Set a pixel-count cap based on memory/performance testing.	Engineering - To confirm
Default redaction	Offer both blur and opaque cover; explain opaque cover for sensitive secrets.	Product - Recommended
Default output format	Preserve a sensible compatible format or let the user choose; finalize after transparency and quality tests.	Product/Engineering - Open

Decision	Recommended starting point	Owner / status
Analytics	Launch without analytics or use a privacy-conscious page-level provider with clear disclosure.	Product - Open
Exact browser support matrix	Target current stable Chrome, Edge, Firefox and Safari; publish verified versions.	QA - To confirm

Appendix A. MVP Feature Checklist

Feature	Release priority
Landing page with clear value proposition	Must
Local image selection and validation	Must
Manual rectangular blur redaction	Must
Manual opaque solid-cover redaction	Must
Multiple editable redaction regions	Must
Flattened image export	Must
Metadata cleaning by re-encoding	Must
Image resize with aspect ratio lock	Must
Output format and quality selection	Must
Download result	Must
Privacy explanation and metadata caveats	Must
Responsive editor UI	Must
Undo/redo and zoom	Should
Clipboard image paste	Should
Automatic OCR and face detection	Future
Batch processing	Future

End of specification

This SRS defines the proposed ShareSafe MVP. Any change that introduces image uploads, accounts, server-side processing, automated recognition or persistent storage should trigger a privacy and scope review before implementation.